SN Computer Science (2024) 5:415
https://doi.org/10.1007/s42979-024-02773-w

REVIEW ARTICLE

Emerging Trends in EEG Signal Processing: A Systematic Review

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Received: 7 November 2023 / Accepted: 5 March 2024

Abstract

This review investigates cutting-edge electroencephalography (EEG) signal processing techniques, focusing on noise reduction, artifact removal, and feature extraction. The study also explores emerging trends such as graph signal processing (GSP), deep learning-based methods, and real-time processing, highlighting their potential in enhancing EEG signal analysis accuracy and efficiency. This research extensively reviews state-of-the-art EEG signal processing techniques and advanced feature extraction methods. Approaches in time, frequency, and time-frequency domains are examined, with applications in cognitive neuroscience, brain–computer interfaces, and clinical diagnostics. The study also explores novel methods like GSP and deep learning, analyzing their impact on EEG signal analysis. The paper presents a comparative analysis of existing methodologies, identifying research gaps and future directions. It emphasizes the significance of GSP in exploring intricate brain networks and dynamic interactions. These findings enhance understanding of brain communication, offering insights into neurological disorders and cognitive functions. Advanced techniques showcased in this study address challenges related to non-stationary and noisy EEG signals, significantly improving accuracy and efficiency in EEG signal analysis. In summary, this review underscores the vital role of EEG signal processing in unraveling the complexities of the human brain. The study's emphasis on robust algorithms and exploration of innovative methods advances EEG signal analysis. This research sets the stage for future developments, fostering progress in the field of EEG signal processing.

Keywords EEG signals · Signal processing · Feature extraction · Classification models · Graph signal processing

Introduction

Electroencephalography (EEG) is a diagnostic technique employed to study electrical brain activity. It is widely used for data analysis, evaluating time and frequency series by measuring voltage fluctuations resulting from ionic currents in neurons [1]. This spontaneous electrical activity is recorded using multiple scalp electrodes, forming the EEG signal. Unlike other neuroimaging methods such as PET, MEG, fMRI, and TMS, EEG stands out due to its affordability, flexibility, excellent temporal resolution, non-invasiveness, user friendliness, portability, and safety. EEG captures changes from synaptic transmissions; when action potentials reach axon terminals, neurotransmitters are released, leading to either excitatory or inhibitory post-synaptic graded potentials at postsynaptic cells [2, 3]. These potentials create ionic currents, generating local field potentials. Synchronized activity among specific pyramidal neurons in cortical brain regions allows the detection of electric fields. The stable orientation of these neurons' currents prevents cancellation, resulting in a significantly stronger overall electric field [4]. This summation process facilitates the measurement of EEG signals, which exhibit nonlinear and non-stationary characteristics. Detecting neuronal activity in these signals relies on the observation of small voltage fluctuations, which can vary based on the expertise of the examiner. However, visual inspection becomes time-consuming for long EEG recordings and may lead to inaccuracies due to signal artifacts.

To overcome challenges and attain quicker and more accurate results, computer-aided technologies are utilized to process and analyze EEG signals. The utilization of these technologies in EEG signal analysis has become widely popular, particularly in diagnosing various neurological and neuropsychiatric disorders such as epilepsy [5, 6], major depressive disorder (MDD) [7, 8], alcohol use

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Published online: 09 April 2024

disorder (AUD) [9, 10], and dementia [11, 12] (including Alzheimer's disease, mild cognitive impairment, Parkinson's disease, and dementia with Lewy bodies). Additionally, EEG's application in Motor Imagery has paved the way for advancements in neuroprosthesis [13]. Moreover, in various research domains, physiological data like EEG has proven to be invaluable. These areas encompass identity authentication [14, 15], sleep stage classification [16], emotion recognition [17, 18], eye state detection [19, 20], and drowsiness monitoring [21]. All of these fields benefit significantly from the valuable insights derived from EEG data analysis. The EEG signal processing and analysis typically involve four essential steps. Initially, raw signals undergo preprocessing, wherein techniques such as filtering are employed to enhance their quality. Following this, critical information is derived from the preprocessed signals in the form of features. These features are then refined using selection methods to enhance their optimization. In the subsequent analysis phase, diseases are diagnosed or various functional brain states are identified utilizing machine learning models or statistical tests. The primary aim of this research study is to thoroughly explore the multitude of EEG-based research applications present in the existing literature. Figure 1 illustrates the step-by-step process outlined in different sections of this paper. Subsequently, a comparative analysis will be conducted to achieve the following objectives:

- Research focusing on varied methods of data collection, including publicly available EEG datasets and locally conducted datasets.
- An array of pre-processing techniques, like down sampling, handling artifacts, and scaling features, is employed.
- Different types of features acquired using a variety of methods for feature extraction.
- The application of post-processing techniques involves methods like feature selection and dimensionality reduction.
- Employing result analysis methods, such as classification algorithms and statistical tests.

EEG Signal

Employing metal electrodes attached to the scalp, an EEG, continually captures the brain's electrical activity. Even during sleep or relaxation, neuronal cells continue to generate electrical currents as a natural form of communication. The basic diagram of EEG signals analysis and classification using different techniques are presented in Fig. 2.

Historical Context

Richard Caton pioneered the capture of electrical signals from human brain activity using a galvanometer and two electrodes in 1875, marking the inception of EEG and its

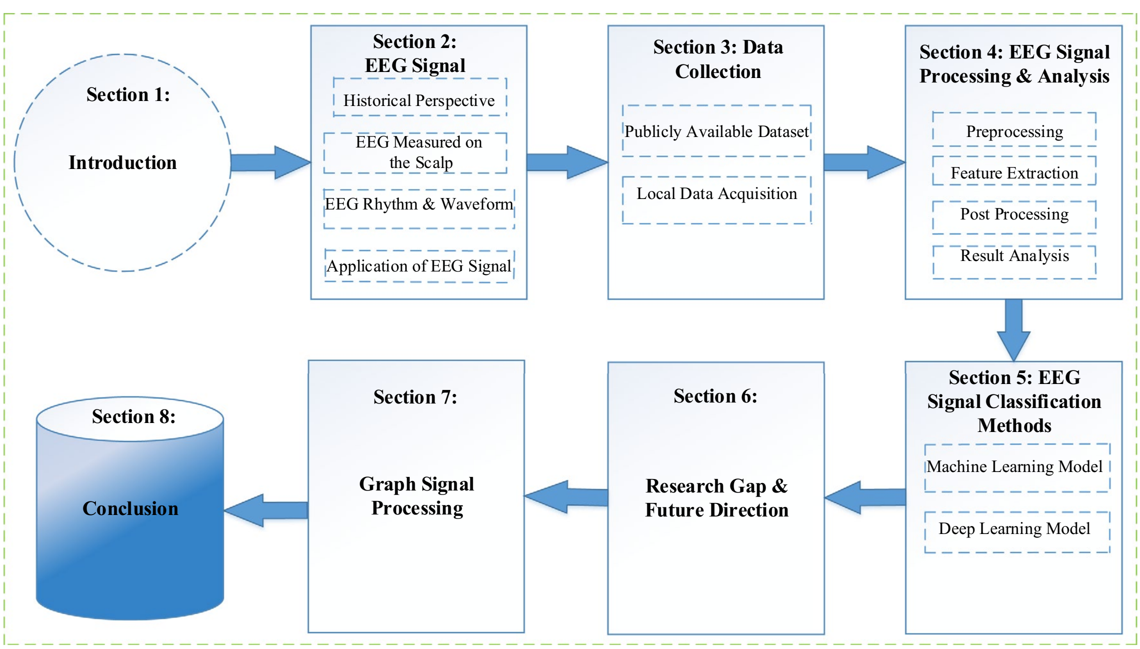

Fig. 1 Overview of the review paper

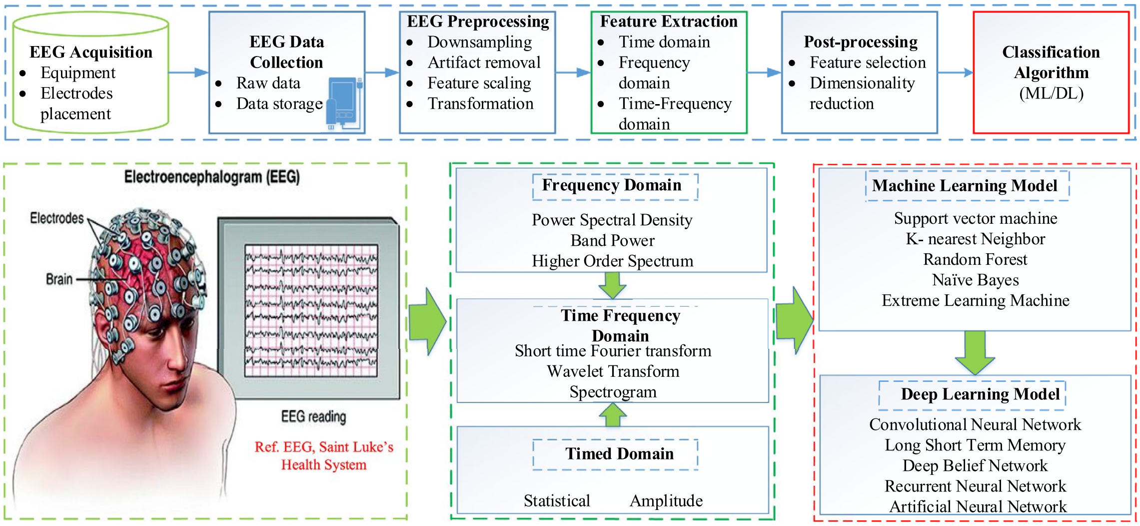

Fig. 2 Diagram providing an overview of feature extraction and classification utilizing EEG signals

applications in illustrating the brain's electrical neural activity. Later, in 1920, Hans Berger identified human EEG waves and employed a string galvanometer. His research, highlighted in studies by [22, 23], revealed the significance of the alpha rhythm and the alpha blocking response within EEG signals. Berger made significant contributions to EEG research during the 1930s, including the first recordings of sleep spindles, descriptions of the impact of hypoxia on the brain, diffuse and specific brain diseases, and descriptions of epileptic discharges. Additionally, he looked on cerebral localization, especially in relation to brain malignancies, and found important links between mental activity and variations in EEG signals. Since the early 1930s, the history of EEG has been a constant and developing process that has resulted in significant advancements in experimental, clinical, and computational endeavors. The progress in EEG technology has significantly improved the understanding, identification, detection, and treatment of various physiological and neurological abnormalities in the brain and central nervous system (CNS). These advancements have enabled the acquisition of EEG signals through both invasive and noninvasive methods, utilizing fully automated computerized systems [23].

Human Brain

The cerebrum is anatomically divided into three major sections: the hindbrain, midbrain, and forebrain, each further segmented into two hemispheres. According to [24], the outer layer of the cerebral cortex comprises the frontal, occipital, parietal, and temporal lobes, forming the four main lobes (refer to Fig. 3). The central nervous system integrates and processes sensory data before transmitting it to higher brain functions, which are in charge of managing

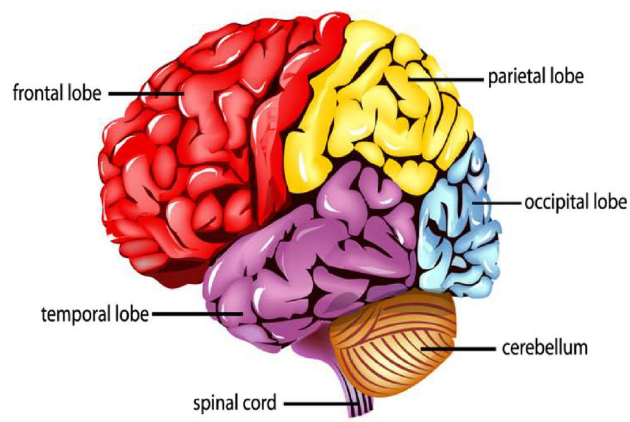

Fig. 3 Different lobes of the human brain [24]

numerous physical functions. Although these functions are spread across different areas to a considerable extent, there is some notable localization of functions worth mentioning.

For example, the frontal lobe is responsible for personality, emotions, and complex cognitive functions. The temporal lobe is essential for processing auditory and other sensory information. In contrast, the parietal lobe is primarily involved in language processing, attention, and sensory perceptions. On the other hand, the occipital lobe plays a vital role in the realm of vision [25]. EEG electrodes are typically placed on the scalp following the globally accepted 10–20 system. This system relies on anatomical points at the back of the skull (inion) and the tip of the nose (nasion). It assigns areas like frontal, parietal, central, temporal, occipital, and auricle using letters F, P, C, T, O, and A, respectively.

EEG Rhythms and Waveform

Scalp recordings capture signals with frequencies below 40 Hz and amplitudes ranging from microvolts to nearly 100 μV. EEG rhythms are traditionally categorized into five frequency bands, and their interpretation varies based on the individual's mental state and age. For example, an adult's EEG significantly contrasts with that of a newborn, featuring notably lower frequencies. This understanding of EEG frequency bands is essential for analyzing brain activity and interpreting EEG data in various contexts. The EEG rhythm [26, 27] encompasses a frequency range between 0.5 and 40 Hz, which is further divided into four main waves: δ, θ, α, β and γ, wave, as depicted in Table 1. Below, the unique features of EEG signals of different frequencies are demonstrated:

1. Delta (δ) band (0.5–4 Hz): The δ band is observed in both the temporal lobe and parietal lobe of the brain. It is characterized by substantial amplitude and is commonly present during deep sleep in adults or when the brain experiences hypoxia. Additionally, the delta band is the predominant waveband in the brains of infants.
2. Theta (θ) band (4–8 Hz): The θ band primarily manifests during adolescence and is associated with emotions and mental states in individuals. It becomes particularly noticeable when adults experience negative emotions.
3. Alpha (α) band (8–14 Hz): The α band, the highest frequency wave in EEG signals, is typically observed at the posterior regions of the brain on both sides. This waveband is often present when the brain is awake or in a relaxed state with closed eyes, making it a fundamental component in EEG signals.
4. Beta (β) band (14–30 Hz): Both sides of the brain mostly show the β band, showing increased excitability of the cerebral cortex. It often denotes increased neuronal activity in the cerebral cortex and is connected to excessive mental stress.
5. Gamma (γ) band (>30 Hz): The γ band, observed in the sensory cortex, represents high-frequency brain waves associated with sensory perception and cognitive processing.

Table 1 EEG signals waveform and their state

Frequency band	Speed (Hz)	Mental state
Delta	0–4	Deep sleep
Theta	4–8	Drowsy, Light sleep
Alpha	8–12	Relaxed
Beta	12–30	Active thinking, Alert
Gamma	30+	Hyperactive, High alert

Different Application of EEG Signals

Electroencephalography (EEG) signals are widely utilized across various disciplines, providing valuable insights into brain activity and cognitive processes. Its non-invasive nature and high temporal resolution render EEG a versatile tool for studying the brain's electrical activity. EEG is a vital diagnostic tool for locating various neurological conditions and abnormalities in the human body. Some key applications of EEG signals include:

- EEG is used for monitoring alertness, coma, brain death, stroke, and tumor providing critical information about the brain's functional state in patients.
- Monitoring cognitive engagement is possible through the analysis of the alpha rhythm, helping assess a person's attention and mental focus.
- In anesthesia, EEG assists in controlling the depth of anesthesia (servo anesthesia), ensuring optimal levels during medical procedures.
- It plays a vital role in investigating epilepsy and locating the origin of seizures, aiding in the diagnosis and management of epileptic conditions.
- EEG is useful in testing the effects of epilepsy drugs, evaluating their impact on brain activity and seizure control.
- In assisting experimental cortical excision of epileptic focus, EEG helps identify the specific brain area responsible for seizures, guiding surgical interventions.
- EEG is valuable in monitoring brain development, especially in infants and children, providing insights into neurological maturation.
- It is employed in testing drugs for convulsive effects, helping assess the potential impact of medications on brain activity.
- EEG is a valuable tool in investigating sleep disorders, investigating mental disorders and understanding the physiology of sleep, shedding light on various sleep-related conditions.
- EEG can be integrated into hybrid data recording systems, combined with other imaging modalities, such as fMRI or PET, to gain a comprehensive understanding of brain function in various contexts.

Data Collection Studies

Various research endeavors employing EEG signals have utilized publicly available datasets or privately generated datasets that remain unreleased to the public. As a result, the data collection process can be divided into two groups in particular: publicly accessible datasets and alternative approaches for gathering data locally.

Publicly Available EEG Datasets

Publicly available datasets are collections of EEG (electroencephalogram) signals that researchers can access for their studies. These datasets are made freely accessible to the public and serve as valuable resources for various research applications. Table 2 provides a comprehensive overview of publicly available datasets. It includes the dataset's name, the number of classes used to distinguish EEG signals. Furthermore, the last column provides insight into the objectives of diverse research investigations that have utilized these datasets for different applications.

Local Data Acquisition Studies

Table 3 provides insights into various research studies where researchers have independently created their own EEG data although these database are not publicly accessible. The table includes information on the purpose behind collecting EEG data, details about the subjects involved, the number of classes, and the EEG channels used for data acquisition. Some references to the respective research studies are provided. These studies cover a diverse array of applications, such as dementia research, depression analysis, motor imagery investigations, emotion and alcoholism exploration etc.

EEG Signal Processing and Analysis

Four phases are involved in EEG signal analysis: preprocessing, feature extraction, postprocessing, and result analysis [43]. The basic flowchart of analysing the EEG signals is presented in Fig. 4. In a deep neural network, there's no need for feature extraction and selection [44]. it means raw signals are directly passed for classification and analysis purpose to the result analysis phase. It means we can ignore the first three stages of the preprocessing mechanism. Also, one can avoid these steps if we use already preprocessed data, mainly some publicly available datasets.

Preprocessing

The pre-processing is divided into three steps i.e. downsampling, artifact removal and feature scaling.

Downsampling: The recorded data by the metal electrode over the scalp with different EEG devices is down samples

Table 2 Publicly available dataset

Ref.	Dataset	No. of classes	Subjects	Sampling rate (in Hz)	Purpose
[28]	Bonn University database	3 (preictal, interictal, ictal)	10	173.61	Epilepsy seizure detection
[29]	Bern Barcelona database	2 (Focal & Non Focal)	5	512	Epilepsy seizure classification
[30]	Florida State University	2 (HC & AD)	24	128	Alzheimer detection
[31]	CHB-MIT Scalp EEG Dataset	2 (seizure and non-seizure)	23	256	Epilepsy seizure
[32]	DEAP database	4 (valence, arousal, dominance, liking)	40	512	Emotion classification
[33]	SEED_EEG dataset	3 (positive, negative and neutral)	15	–	Emotion detection
[34]	Keirn and Aunon dataset	5 (mental activities)	7	250	Mental task classification
[35]	EEGMAT MIT Physionet	2 (mental activities)	36	500	Mental task identification
[36]	BCI competition II dataset-III	2 (Left or Right hand movement)	1	128	Hand movement
[37]	UCI dataset for Alcohol	2 (control and alcohol)	122	256	Alcohol detection

Table 3 Some previous research on local datasets (Not available publicly)

Ref.	Objective	Subjects	No. of class	Channels
[10]	Alcoholism	HC:30, AUD:30	2	19
[21]	Driver drowsiness	HC:13	1	9+2 earlobe channel
[38]	Dementia (Alzheimer, MCI)	HC:37, MCI:37, AD:37	3	19
[39]	Depression, stress	HC:13, MDD:13	2	30 + avg (M1, M2) as reference
[40]	Motor imagery	HC:15	1	61
[41]	Emotion recognition	HC:15, MDD:15	2	16
[42]	Eye state detection	HC:2	1	1

to different frequency such as from 512 to 64 Hz, 256 to 16 Hz depending upon the requirement.

Artifact removal: Artifacts are generated due to errors such as experimental settings, environmental noise, physiological signals, or biological artifacts. These include eye movements, muscle activity, and ECG artifacts. We should remove these artifact types to get optimized results [43].

Feature scaling: Scaling is crucial for the dataset's features to display symmetrical behavior. One of the most commonly used feature scaling method is normalization. Some machine learning algorithms' objective functions won't function effectively without normalization since the range of raw data's values fluctuates wildly.

Feature Extraction

Feature extraction involves extracting features from the primary signal to obtain consistent categorization. Dimensionality Reduction using feature extraction is the problem of finding the dimensions along which the dataset shows the most significant variability, which corresponds to the most relevant parameters of the dataset. Feature extraction, dimension reduction, or feature selection are used to minimize the initial collection of features if each feature impacts categorization [45]. Meaningless or redundant features are removed during feature selection/reduction [46], resulting in faster performance and classification compared with the initial features. Feature extraction is instrumental in classifying neurological disorders and other monitoring applications using EEG signals. As EEG signals contain many data points, distinct and meaningful features are recovered using various feature extraction methods [47, 48]. The most common method such as wavelet transform (WT) [49], power spectral density (PSD), statistical, short-time Fourier transform (STFT) [43], wavelet entropy (WE), differential entropy (DE), empirical mode decomposition, etc. Frequencies and amplitudes can represent signal patterns in EEG signal analysis [50].

Non-parametric approaches for delivering spectral estimates are based on power spectral density (PSD) descriptions [51]. The periodogram [52] and the correlogram are two well-known nonparametric techniques. It gives a viable high resolution for sufficiently prolonged data lengths. The Eigenvector methods estimate the frequencies [48] and powers of signals from noise-corrupted data and derive from the Eigen decomposition of the noise-corrupted signal's correlation matrix [53]. Different feature extraction techniques for EEG are given in Table 4. The feature extraction phase reduces the initial data into a lower dimension, containing the initial vector's practical information. As a result, based on the nature of the collection, it is critical to identify the essential elements that characterize the entire dataset.

Time Domain Analysis

Time domain analysis of EEG signals involves examining the signal characteristics as they evolve over time. This includes understanding the amplitude, duration, and shapes of various EEG waveforms.

- Amplitude Analysis: Measures the intensity or strength of the EEG signal. High amplitude might indicate strong neural activity, while low amplitude can suggest relaxed or inactive brain states.
- Temporal Patterns: Involves identifying recurring patterns in the EEG signal, which can indicate regular brain activities or abnormal patterns like seizures.
- Statistical Measures: Variability, mean, standard deviation, skewness, and kurtosis are statistical measures applied to EEG segments to understand their distribution and characteristics.

Frequency Domain Analysis

Frequency domain analysis aims to reveal the underlying frequency components within EEG signals. Different brain activities manifest in different frequency bands.

- Power Spectral Density (PSD): PSD illustrates how signal power is distributed across various frequencies. It's calculated using techniques like Fast Fourier Transform (FFT) and provides insights into dominant frequency bands.
- Band Power: Analyzing the power within EEG frequency bands can provide specific information about the brain state.

Time-Frequency Domain Analysis

EEG signals are dynamic and display shifts in frequency over time, highlighting the importance of time-frequency analysis.

- Short-Time Fourier Transform (STFT): STFT enables us to observe the alterations in the frequency content of EEG signals within brief time intervals. By using windows of varying sizes, it provides a time-localized frequency analysis, making it suitable for non-stationary signals.
- Continuous Wavelet Transform (CWT): CWT analyzes signals at different scales and provides high resolution in both time and frequency domains. It's particularly useful

Table 4 Summary of feature extraction technique

Feature extraction technique	Feature name	Description/Advantages	Disadvantages
Spectral estimation methods [54, 55]	Nonparametric or classical approach	Welch method is used	Suffer from spectral leakage effect due to windowing
	Parametric or non-classical approach	Estimating PSD of signals and Graph	Deciding the optimal model order (i.e lower and high order)
Family of transforms [56, 57]	Fourier Transform (FT)	Transform Time domain signals to frequency domain Overcome the problem of spectral leakage and appropriate for stationary signals	Less accurate for non-stationary signals Discontinuity in the signal cannot be represented appropriately
	Short time fourier transform (STFT)	Represents the signals in time and frequency domain It is window version of FT	Higher temporal resolution and lower frequency are the results of shorter time intervals Larger time interval results in lower temporal resolution and higher frequency
	Continuous wavelet transform (CWT)	Function has two parameters a and b,known as "scale" and "translation" respectively Alternate method of transformation to STFT	Parametes, a and b, vary continuously
	Discrete wavelet transform (DWT)	Low and high pass filters are used for analysis purpose of signals at various "scales" "Level wise" decomposition which contains detailed and approximation coefficients respectively	Selecting the accurate "wavelet" is a difficult task Having different issues such as aliasing,lack of directionality and shift variance
Graph signal processing (GSP) [58, 59]	Graph Fourier transform (GFT)	Obtain from eigenvectors of Laplacian matrix of a graph signals Represent signals into vertex and graph spectral domian Depends on Laplacian eigenvector	Identifying the right graph structure Identifying the right frequency representation of graph signals
	Graph wavelet transform (GWT)	Graph signal is localized in vertex and spectral domain Represented as diffusion wavelet and spectral graph wavelet Diffusion is used as a scaling tool	Translation and scaling differs from classical signal processing
	Spectral graph wavelet transform (SGWT)	In SGWT, scaling functions extract information from the spectral decomposition of the Graph Laplacian (L) It is simpler than diffusion wavelet SGWT offers better management on the selection of wavelet scales	Shift sensitivity Lack of direction Insufficient phase information

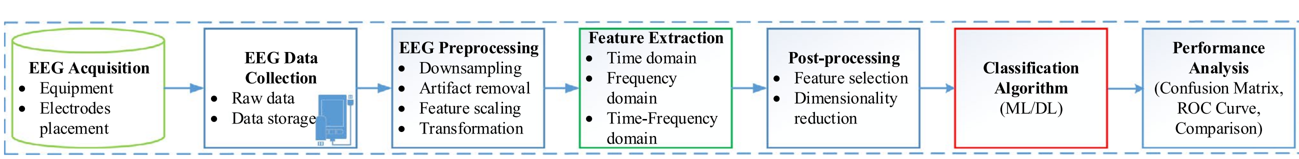

Fig. 4 Fundamental structure for examining EEG signals

for capturing transient events and frequency changes over different time intervals.

- Spectrogram: This graphical representation illustrates how the frequency spectrum of a signal changes over time. Darker areas on the spectrogram indicate higher power or amplitude in specific frequency ranges at specific times.

Postprocesssing

Feature selection and the dimensionality reduction technique are the types of postprocessing [43]. To build a model, one must choose a subset of pertinent features using the feature selection technique. Feature selection removes irrelevant variable or noise, reduces computational complexity and reduces overfitting and increase generalization capacity of the model [60]. Dimensionality Reduction (sometimes also known as feature extraction) combines features to extract a new set (smaller than the starting one) of features. It is a technique used to retain the necessary information while minimising the process time and dimension of the initial feature vector [61]. To interpret the data and obtain more precise results, the dimension of EEG signals needs to be reduced. Furthermore, to improve classification accuracy, the characteristics or dimensions must be decreased. Using feature extraction, data in a higher-dimensional domain is turned into a lower dimensional domain. Data transformations can be linear, like in PCA [62, 63], but nonlinear dimension reduction approaches are also available. To minimize the number of dimensions, LDA [61, 62], PCA [63], and ICA techniques can be used. In order to clarify our grasp of dimension reduction, these three strategies will be defined. Two methods for extracting components from two-dimensional non-Gaussian datasets are PCA and ICA. PCA recovers the components with the greatest variance, whereas ICA recovers the components with the greatest independence.

Result Analysis

The researcher can use various machine learning algorithms [61, 62] such as supervised, unsupervised, deep learning neural networks [64, 65], and GSP [58] techniques for EEG signal analysis. With the help of these techniques and EEG signals, one can diagnose neurological disorders such as epilepsy or seizures and monitor other applications such as emotion detection, sleep stage categorization, etc. Supervised learning is used for classification (discrete) and regression (continuous) and Unsupervised learning for clustering and dimensionality reduction. A machine learning model [61] is defined by a set of parameters implemented by a computer program that uses training data or prior experience to optimize the parameters of models. A supervised machine learning technique is a support vector machine (SVM) [66] that can also use for medical diagnostics. In addition to machine learning, deep learning algorithms are also utilized to predict depressive subjects and brain disorders with EEG signals. LSTM is primarily applied to prediction tasks among all deep learning techniques. For additional chances to obtain the highest accuracy, the author of [64, 65] applied two deep learning architectures to a single prepared dataset.

EEG Signal Classification Methods

The fusion of deep learning and machine learning has significantly improved our understanding of brain activity, especially in EEG signal classification. This process involves stages like preprocessing, where techniques such as filtering and artifact removal refine raw EEG signals. Informative features like time-domain statistics, power spectral density, and spectrograms are extracted, and machine learning models like support vector machines and random forests are employed for classification. These methods have shown effectiveness in achieving accurate results. Additionally, the introduction of deep learning models, particularly CNNs and LSTM, has revolutionized EEG signal classification. CNNs are adept at automatically learning hierarchical spatial patterns in EEG data, leveraging multiple convolutional layers for feature extraction and recognition. RNNs, on the other hand, excel in capturing temporal dependencies within EEG sequences, making them suitable for tasks where temporal dynamics are crucial. The integration of both CNNs and RNNs in hybrid architectures like ConvLSTM has yielded even more robust results. To prevent overfitting, techniques like dropout, batch normalization, and early stopping are commonly employed. Integrating machine learning and deep learning has greatly improved EEG signal classification. By merging preprocessing, feature extraction, and classification stages with traditional algorithms and deep learning models, accuracy has increased, providing a better understanding of intricate brain activities. Different features extracted from EEG data using various methods undergo classification through suitable algorithms in the analysis of EEG signals. The most prevalent classification algorithms encompass KNN [67], SVM [68], naive Bayes (NB) [69], logistic regression (LR) [70], gradient boosting (GB) [71], LDA [72], RF, and DT. Additionally CNN, deep DNNs [73], deep belief networks (DBN) [74], ANN [75], and recurrent neural networks (RNN) possess capabilities for emotion identification. Consequently, recent investigations have frequently employed deep learning classification techniques for emotion estimation.

Deep learning [76] is a subset of machine learning that employs artificial neural networks to discern intricate patterns and correlations within datasets. This sophisticated approach involves the utilization of multiple layers in neural networks, allowing the model to automatically extract hierarchical features from the input data. The depth of these networks enables them to learn complex representations, making deep learning particularly effective in tasks such as image and speech recognition, natural language processing, and other domains where understanding intricate relationships in data is crucial.

Enhanced performance is observed with spiking neural networks and hierarchical fusion CNN compared to standard CNN. In machine learning performance analysis, essential techniques include the Confusion Matrix, ROC Curve, and Model Comparison. The Confusion Matrix categorizes outcomes into True Positive, True Negative, False Positive, and False Negative, providing insights into a model's accuracy. The ROC Curve illustrates sensitivity and specificity trade-offs, crucial for binary classification. AUC quantifies a model's ability to differentiate between classes. Model Comparison involves evaluating models using metrics like accuracy, precision, recall, F1-score, or AUC, essential for selecting the best model. These techniques are vital for assessing and enhancing machine learning model accuracy, guiding data-driven decisions.

In the realm of EEG signal processing, a comparative analysis between traditional methods and newer techniques unveils distinctive strengths and limitations inherent in each approach. Traditional methods, such as time domain and frequency domain analyses, have long been the bedrock of EEG signal processing. Time domain analysis provides a well-established framework but falls short in capturing complex spatial interactions and nonlinear patterns within EEG signals. Frequency domain analysis, including techniques like FFT and wavelet transform, offers insights into signal frequency components, yet it tends to overlook the intricate spatial relationships between EEG channels, limiting the capacity to discern network dynamics. On the other hand, newer techniques rooted in GSP [58] present innovative avenues for EEG analysis. Graph theory representation, where EEG channels are nodes and their connections are edges, captures spatial dependencies effectively. The graph Fourier transform enables the exploration of EEG signals in the graph frequency domain, allowing for the extraction of localized and distributed information. Graph Convolutional Networks (GCNs) [58, 59] apply convolutional operations directly on graph-structured data, excelling in capturing spatial dependencies. In comparing these approaches, several key factors emerge. Traditional methods often lack explicit spatial information and struggle with capturing complex spatial dependencies. In contrast, newer graph-based methods excel in representing spatial relationships, adapt well to non-linear patterns, and offer a more comprehensive view of network dynamics. However, newer techniques may introduce added complexity and potential interpretability challenges, especially in cases where large amounts of labeled data for training are required. Ultimately, the choice between traditional and newer graph-based techniques hinges on the specific research question, dataset characteristics, and the desired balance between interpretability and complexity.

Research Gaps and Future Direction

One of the essential questions is how to represent efficiently, handle, analyze, and visualize large-scale structured data, particularly data from complicated domains like networks and graphs, which is one of the significant challenges in existing machine learning techniques. GSP [58], a developing branch of signal processing models and methods aimed at estimating data based on graphs, offers a new route for study to address this issue. In this article, we examine how GSP concepts and tools, including graph filters and transforms [59], have aided the development of cutting-edge machine learning approaches. More importantly, we offer a new perspective on the future expansion of GSP development [58, 59], which could act as a link between applied mathematics and signal processing on the one hand, and machine learning and social network analysis on the other. The features from the EEG signal obtained in the previous stage are then extracted and turned into a feature vector. There are some problems and issues that arise while going through the analysis of EEG signals. The following are the most common:

1. Most applications have lacked access to larger datasets, limiting the validation of the models for use in real-world applications and the viability of using them in clinical practice.
2. Because real-time data is rife with artefacts, highly efficient filtering approaches must be proposed to help decrease noise and improve model performance.
3. For various datasets, a range of feature extraction approaches and machine learning, deep learning classifiers such as CNN and LSTM can be implemented in graph domain.
4. To better comprehend the complexity of EEG signals, non-linear and functional connectivity aspects can be examined for a variety of applications.
5. Traditional signal processing methods face difficulty and take high computational time to deal with irregularly structured data.
6. A recent trend in signal processing, termed "Graph Signal Processing" (GSP), is developing rapidly and deals with irregularly structured data.

7. GSP has become a natural choice for researchers to deal with neurological disorders, such as alzheimer, epilepsy, brain image analysis, and EEG signal applications.

Apart from above discussion about some limitation and research gap which can be dominated by using the one of the recent trend technique called as GSP. Difficulty attain in point number 4 and 5 is well addressed by recent technique called as GSP.

Challenges such as noise and artifacts, irregularly structure data, large datasets, limited spatial resolution, and individual variability necessitate innovative approaches. The gaps in EEG signal processing arise from a combination of inherent limitations in the data, such as noise and resolution constraints, as well as the complex and variable nature of the human brain. Addressing these gaps requires innovative approaches and methodologies, including advanced signal processing techniques like GSP, to enhance the reliability and applicability of EEG-based research and clinical applications. GSP offers a promising avenue to overcome these hurdles. By representing EEG data as a graph and leveraging GSP techniques, researchers can capture spatial relationships, enabling a more nuanced understanding of neural connectivity. GSP can also facilitate multimodal integration, combining EEG with other imaging modalities to create a comprehensive view of brain activity. Adaptive filtering within the GSP framework enhances noise removal, contributing to cleaner and more accurate data.

Graph Signal Processing for EEG Signals

Graph signal processing (GSP) [59] is widely utilized in brain image processing due to its effectiveness in analyzing irregular domain data and graph networks. In neuroscience, complex brain activities are associated with intricate functional and structural connectivity networks. GSP proves invaluable for managing such irregular and dynamic non-Euclidean data, commonly found in applications like social networks, sensor feeds, online traffic, supply chains, and biological systems. Traditional deep learning methods face limitations in handling irregular domain data. When data lacks a Euclidean structure, standard deep learning algorithms struggle. If we want to utilize the CNN on the graph structure [77, 78] or data with irregular domain, the CNN not perform well and it will be beyond the scope of CNN analysis. Geometric deep learning is a field of research that extends deep learning to non-Euclidean data. To take into account the merits of GSP, a classifiers GCNN (graph convolutional neural network) have been developed in [77, 78].

But one limitation of GSP is the computational complexity associated with analyzing signals on large and complex graphs. As the size of the graph increases, the computations required for filtering and processing signals become more demanding, posing challenges for real-time applications and efficient algorithm implementations.

Graph Convolutional Network (GCN)

CNNs cannot be directly utilized on non-Euclidean structured data due to the limitations of fixed convolution kernels, which cannot capture all the node neighborhood information. For the non-Euclidean structured data GCNN [77] is applied on the EEG signals as graph vertices. Convolution and pooling operations on graph are performed by using GCNN model [77, 79] based on GSP approach. Graph signal can be represented by $G = \{V, E, W\}$ with N vertices and also defined as an undirected and weighted graph. Here V is known as set of vertex, E known as set of edges, and W defined as $N \times N$ matrix grid storing edge weights. When all N vertices are considered in GSP, a signal $x{:}V \to \mathbb{R}$ is characterised on vertex and can be addressed with a vector $\mathbf{x} \in \mathbb{R}^N$. The equation $\tilde{\mathcal{L}} = I_N - \mathbf{D}^{-1/2}\ \mathbf{W}\mathbf{D}^{-1/2}$ is referred to as the normalized graph Laplacian [79, 80] where I_N is an identity matrix of size N and $\mathbf{D}$ represents a diagonal matrix. The normalized graph Laplacian's range is defined by its Eigen decomposition, $\tilde{\mathcal{L}} = \mathbf{U}\Lambda\mathbf{U}^{\mathbf{T}}$, where the columns of U comprise of eigenvectors $\{u_l\}, l = 1, \ldots, N$, and Λ is a diagonal matrix with graph frequencies $\{\lambda_l\}, l = 1, \ldots, N$, as the diagonal elements. The graph Fourier transform of graph signal $\mathbf{x}$ on graph G is expressed as $\tilde{\mathbf{x}} = U^T$x while the inverse graph Fourier is defined as $\mathbf{x}= \mathbf{U}\tilde{x}$ [79]. The detail explanation of Laplacian, graph Fourier transform and Chebyshev polynomial is given in [77, 79]. The convolution of irregular data is implemented by the graph convolution network. The graph convolution network incorporates spectral-based [77] and space-based graph convolution [78]. Some graph neural network classifiers based on graph signal processing are presented in Table 5.

Conclusion

This research explores diverse EEG applications, comparing data acquisition methods, and scrutinizes signal processing techniques: preprocessing, feature extraction, and post-processing. Emphasis lies on result analysis, focusing on classification through machine learning models. Each stage significantly contributes to processing raw EEG signals, enabling automated decision models. This paper provides an overview of EEG signal analysis techniques, including different transforms and machine learning methods for classification. The study also discusses challenges in EEG analysis and explores the application of concepts like GSP and GNN to address these challenges effectively.

Table 5 Summary of graph neural network based classifiers

Method	Objective	Merits	Demerits	Ref.
GCNN	• Practical applications of GNN include human behavior detection, Video identification and biological network	• GCNs are an extended form of CNNs, designed to operate on data with complex non-regular structures. structures	• Less handling of edges of graphs based on their types and relations	[79]
DGCNN	• Emotion recognition using dynamic graph convolutional neural network	• Consist discriminatory EEG feature extraction and intrinsic relation between EEG signals Alleviate the limitations of the GCNN	• Data scales of the EEG databases relatively small Much larger scales is desired for performance	[80]
RGNN	• Emotion recognition using regularized graph neural network	• Find the link between emotion related brain regions and channels. Give better performance as compared to previous literature	• Used only two regularizers for capturing the changes in EEG. How to implement on smaller number of channels	[81]
LGCN + Focal loss	• Linear graph convolution network with focal loss is used to detect seizure	• Performance is much better than compared with the two imbalance learning method	• This method more focus on the minority seizure samples while less focus on majority non seizure sample	[82]
TGCN	• Used for automatic seizure detection with Temporal graph convolution networks	• Extract spatio-temporal features with localization and sharing, Obtain brain affected regions and timing of seizures occurring	• Computational time is more	[83]
GCN	• GCN for time-resolved EEG motor imagery signals	• Obtain Pearson coefficients to build the Laplacian matrix. Features achieve relatively high accuracy in a short time. Applied on irregular structure data	• Less informative data	[84]

The rise of GSP signifies a breakthrough in comprehending graph-structured biological data. Our literature review concentrated on analyzing biological data within the realm of graphs, encompassing GSP, GNN, and graph learning methods. Despite existing challenges, these graph-based approaches hold substantial potential in the fields of biological engineering and science.

Exploring the potential implications for future research and clinical practice of EEG signals holds significant promise. In the realm of research, further investigations could delve into refining signal processing techniques, enhancing spatial resolution, and exploring advanced machine learning algorithms for more accurate and nuanced interpretations. On the clinical front, harnessing EEG signals may lead to the development of personalized treatment plans, improved diagnostic tools, and real-time monitoring of neurological disorders. The integration of EEG data with other modalities could also open new avenues for comprehensive healthcare approaches. Ultimately, continued exploration and innovation in EEG signal analysis have the potential to revolutionize both research methodologies and clinical interventions in the field of neuroscience.

Author Contributions Ramnivas Sharma: Original draft, Software, Review and editing of the paper, Formal analysis, and Results obtained. Hemant Kumar Meena: Supervised, cross-checked, and edited.

Data availability Publicly available data has been referred to in this paper.

Declarations

Conflict of interest No competing financial or interpersonal conflicts.

Ethical approval Not involve any studies with animals or humans.

Informed onsent Human participants are not involved.

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